14.4 Determining Possible Associations in Data

Lesson Introduction

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<tr><td>Benchmark</td><td colspan="2">MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies, and determine possible associations in terms of a real-world context.</td><td>Learning Target</td><td colspan="2">I can determine possible associations from a two-way table and segmented bar graph.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td colspan="3">N/A</td></tr>
<tr><td>Essential Question</td><td colspan="5">How can you determine possible associations from a two-way table and segmented bar graphs?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">Students build from the previous lesson where they interpreted data from segmented bar graphs. In this lesson, students determine possible associations from a two-way frequency table and segmented bar graphs.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>14.4.1 Warm-Up
(5 min.)</td><td>Students complete and interpret a two-way frequency table
(SE p. 321)</td><td>14.4.3 Bring It Together
(15 min.)</td><td colspan="2">Using segmented bar graphs to find possible associations (SE p. 323)</td></tr>
<tr><td>14.4.2 Guided Instruction
(15 min.)</td><td>Finding possible association in data
(SE p. 322)</td><td>14.4.4 Try It
(10 min.)</td><td colspan="2">Practice finding a possible association from two-way frequency table and segments bar graphs
(SE p. 323)</td></tr>
<tr><td>14.4.5 Wrap-Up
(5 min.)</td><td colspan="4">Quadrant self-reflection activity (SE p. 324)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
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Association
<u>Additional Terms:</u> N/A</td><td colspan="2">N/A</td><td>N/A</td></tr>
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<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may think there is an association between tree type and tree height in 14.4.4 Try it.
Students may struggle with the fractions in 14.4.3 Bring It Together when referring to the amount that is orange in the segmented bar graph.</td><td>Some students may prefer using the two-way frequency table to using the segmented bar graphs.
The definition for association was also defined in Unit 3, Lesson 5, but only specified for numerical data.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
14.4.2 Guided Instruction utilizes the Think-Pair-Share routine, question 1a is the individual Think portion, question 1b is the Pair portion. As a whole class, to hear the thinking and reasoning of others in the class, consider having pairs of students to share whether they agree more with Orlando or Hannah.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
14.4.2 Guided Instruction provides students the opportunity to analyze the mathematical thinking of others through two sample students’ thinking.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider having students label the values of joint frequencies in each segmented bar graph, so that students can easier see the relative frequencies as portions of the bars.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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14.4.1 Warm-Up: Playing Instruments and Sports

Component Overview: *14.4.1 Warm-Up provides students the opportunity to spiral learning from Lessons 1 and 2 of Unit 14. Students should complete this independently, so student data can be collected on their understanding of how to complete and interpret data from a two-way frequency table.*

14.4.1 Warm-Up: Playing Instruments and Sports

Orlando asked his classmates these questions:

- Do you play a sport?
- Do you play a musical instrument?

1. The two-way table gives partial results from the survey. Complete the table, assuming that all students answered both questions.

	Plays Instrument	Does not Play Instrument	Total
Plays Sport	5	11	16
Does not Play Sport	5	4	9
Total	10	15	25

2. What fraction of the total students do not play an instrument and also do not play sports?
   The fraction of the total students who do not play an instrument and also do not play sports is $\frac{4}{25}$.

3. Does the fraction you found represent a joint or marginal relative frequency?
   The fraction I found in #2 represents a joint relative frequency.

Use informal observation of those who are struggling versus proficient to inform additional support when students get to 14.4.3 Bring It Together and 14.4.4 Try It.

Avoid asking for possible associations during 14.4.1 Warm-Up, as students will be looking for possible associations during 14.4.2 Guided Instruction.

14.4.2 Guided Instruction: Finding Possible Associations in the Data

Component Overview: *During 14.4.2 Guided Instruction, students will learn about association, then analyze the thinking of others' description of possible associations in the data.*

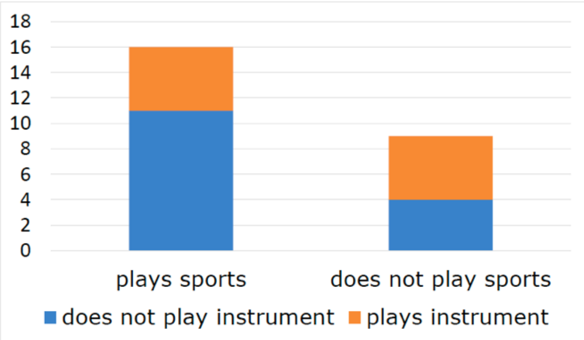
14.4.2 Guided Instruction: Finding Possible Associations in the Data

Two-way tables are effective for organizing data. Creating a segmented bar graph for the data in a two-way table can help make possible associations more visible.

An ***association*** is a way to describe the relationship between the two variables in a bivariate data set. For categorical data, descriptions include strong or weak.

Orlando would like to know if there is any association between playing an instrument and playing a sport. He makes a segmented bar graph from the data in the two-way table.

	Plays Instrument	Does not Play Instrument	Total
Plays Sports	5	11	16
Does not Play Sports	5	4	9
Total	10	15	25



Position students as sense makers of the term association by asking them, before moving onto question 1 where they are analyzing the thinking of others, to determine any possible associations they see in the data.

Consider having students label the values of the joint frequencies on the segmented bar graph to have students continue to make connections on how the data in a two-way frequency table is displayed in a segmented bar graph.

1. Orlando and Hannah discussed possible associations in the data using the displays. Each student's thinking is shared in the table.

Orlando's Thinking	Hannah's Thinking
"Of the students who play sports, there are 5 people who play an instrument. This is also true of the students who do not play sports. This means there is no association, or relationship, between playing sports and playing an instrument because it's 5 people in both cases."	*"While there are only 5 people in each category who play an instrument, there is a much larger portion of the students who do not play sports. Half of the students who don't play sports do play an instrument. I think there is an association between playing sports and playing an instrument. It seems more likely a student plays an instrument if they don't play a sport."*

a. Without the help of your partner, explain whether you agree more with Orlando or Hannah.
Answers will vary. Hannah; it is important that a larger portion of students surveyed do not play sports. This makes it important to compare the five students in each group who play an instrument to the marginal frequency for that category. It is just like comparing the two fractions $\frac{5}{16}$ and $\frac{5}{9}$. Even though they both have a 5 in the numerator, their denominators are very different.

b. Compare your answer with your partner's. Make any edits to your explanation based on your discussion.
Answers will vary.

For students who do better when hearing student thinking than reading it, consider selecting two students to be Orlando and Hannah to read their thinking before having students complete the questions that follow.

14.4.3 Bring It Together: Using Segmented Bar Graphs to Find Possible Associations

Component Overview: *Students will summarize the possible association in the data from the segmented bar graph in 14.4.2 Guided Instruction by completing the statements.*

14.4.3 Bring it Together: Using Segmented Bar Graphs to Find Possible Associations

1. Complete the statements.

   a. Segmented bar graphs are useful for displaying categorical ○ univariate ○ bivariate data.

   b. Segmented bar graphs can be helpful for making visual comparisons between joint and marginal frequencies. Take, for example, the bar on the left of the graph from the previous component labeled "plays sports."

   About ○ $\frac{1}{2}$ ○ $\frac{1}{3}$ of this bar is orange. The other bar, labeled "does not play sports" is just over ○ $\frac{1}{2}$ ○ $\frac{1}{3}$ orange. This means a larger portion of the students who ○ play a sport ○ do not play a sport play an instrument, which means there may be an association between playing sports and instruments.

Consider displaying the segmented bar graph for all the students to see when completing the statements. In question 1b, this will be particularly helpful, because it is referred to directly, and by having it displayed, it will help students see what the statement is referring to.

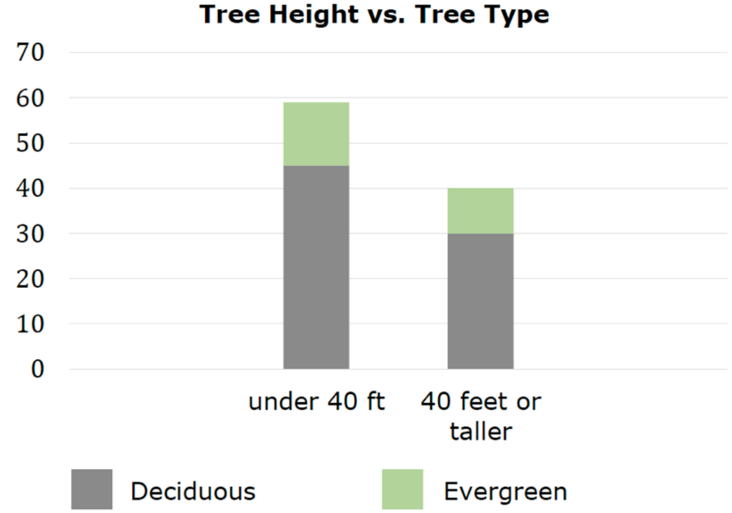
14.4.4 Try It: Possible Associations

Component Overview: *Students will explore how two different, segmented bar graphs help determine if there is evidence of an association in the data.*

14.4.4 Try It: Possible Associations

1. An ecologist is studying a forest with a mixture of tree types. Since the average tree height in the area is 40 feet, the height of each tree is measured against that and recorded. The type of tree is also recorded. The results are shown in the table and segmented bar graphs.

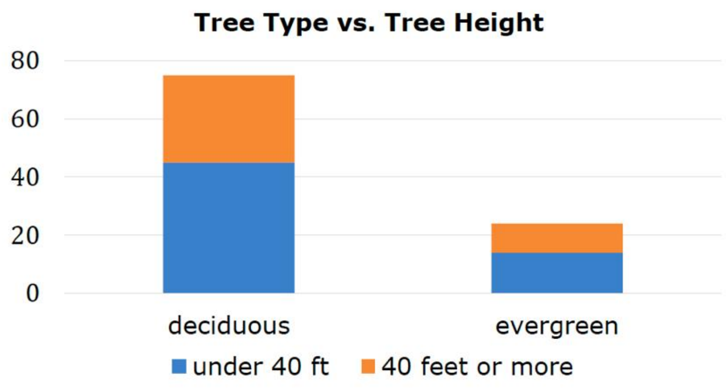
	Under 40 Feet	40 Feet or Taller	Total
Deciduous	45	30	75
Evergreen	14	10	24
Total	59	40	99



To help students analyze similarities and differences of the two segmented bar graphs, consider the following questions:

- *How are the marginal frequencies displayed in each graph?*
- *How are the joint frequencies displayed in each graph?*
- *What is similar about the two graphical displays?*

14.4.4 Try It: Possible Associations (continued)



a. Explain whether or not there is evidence of an association between tree height and tree type.
Answers will vary. Based on the data we are shown, there does not appear to be an association between tree type and tree height. In the Height vs. Type bar graph, about $\frac{1}{4}$ of each height category bar is evergreens, so about the same proportion of trees in each height group were evergreens. In the Type vs. Height bar graph, 60% of the deciduous trees were under 40 feet while 58% of the evergreen trees were under 40 feet, so there isn't much of a difference when the data is sorted by tree type either. Since the proportions of trees in each group are relatively similar even when sorted by different variables, this is good evidence that there is not an association between tree height and type in the given data.

b. Explain which data display was more helpful in determining whether or not there is evidence of an association between height and type of tree.
Answers will vary.

Consider implementing the Stronger and Clearer Each Time instructional routine. Have students think individually about their responses for questions 1a and 1b, then pair students to refine and clarify their responses through conversation, and then finalize their written response.

Facilitate a conversation by prompting students to share their responses to questions 1a and 1b. Have students justify their thinking by referring to the provided segmented bar graphs.

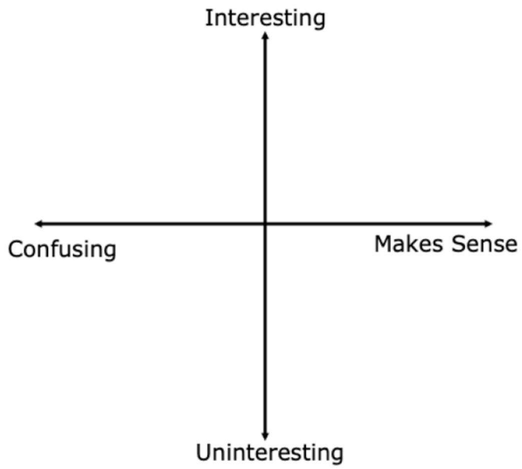
For a challenge, have students determine how the segmented bar graphs would change to show an association in the data.

14.4.5 Wrap-Up: Reflection

Component Overview: *Students complete a self-reflection on the learning target, "I can determine possible associations from a two-way table and segmented bar graphs."*

14.4.5 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate where you are on today's learning target, "I can determine possible associations from a two-way table and segmented bar graph."
   Answers will vary.



Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for students' learning progressions.

Consider our online resources for additional remediation opportunities for students to practice or test their abilities.